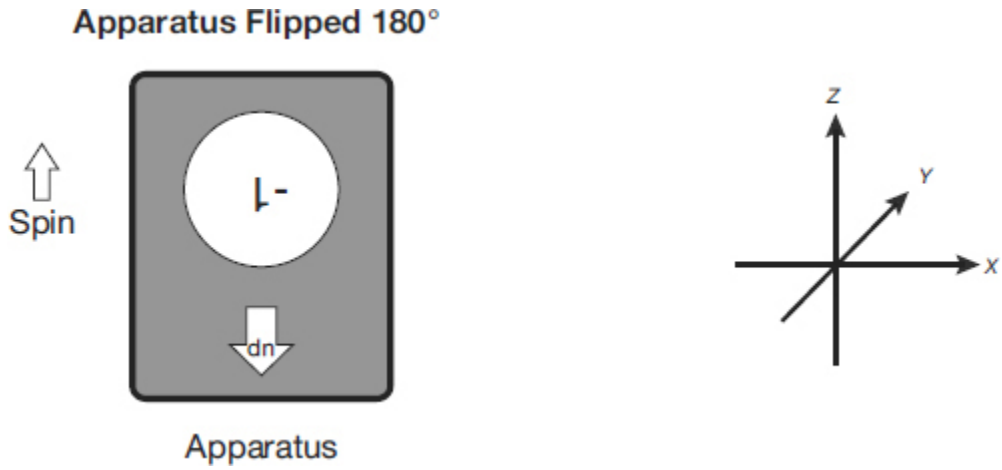


Figure 1.2: The apparatus is flipped without disturbing the previously measured spin. A new measurement results in $\sigma_z = -1$.

Now let's do something new. After preparing the spin by measuring it with $\mathcal{A}$, we turn the apparatus upside down and then measure σ again (Fig. 1.2). What we find is that if we originally prepared $\sigma = +1$, the upside down apparatus records $\sigma = -1$. Similarly, if we originally prepared $\sigma = -1$, the upside down apparatus records $\sigma = +1$. In other words, turning the apparatus over interchanges $\sigma = +1$ and $\sigma = -1$. From these results, we might conclude that σ is a degree of freedom that is associated with a sense of direction in space. For example, if σ were an oriented vector of some sort, then it would be natural to expect that turning the apparatus over would reverse the reading. A simple explanation is that the apparatus measures the component of the vector along an axis embedded in the apparatus. Is this explanation correct for all configurations?

If we are convinced that the spin is a vector, we would naturally describe it by three components: σ_z, σ_x, and σ_y. When the apparatus is upright along the z axis, it is positioned to measure σ_z.